

Fractal Cognitive Space-Time Networks: A Framework for AI-Generated Cognitive Virtual Environments over NeuroDigital Infrastructure

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Abstract

The convergence of artificial intelligence, immersive environments, and neurodigital interfaces is enabling a new class of systems: Cognitive Virtual Environments dynamically shaped by human cognition. In this paper, we introduce Fractal Cognitive Space-Time Networks (FCSTN), a novel framework where virtual universes are generated from mathematical structures including fractal manifolds and relativistic space-time geometries. Within this paradigm, AI agents, neural signals, and distributed computational infrastructures interact to continuously evolve the topology and dynamics of the environment. We propose an architecture that integrates fractal-based procedural generation, AI-driven geometry adaptation, and neurodigital feedback loops, enabling bidirectional coupling between cognition and virtual space. The framework leverages coalition formation mechanisms from Device-to-Device (D2D) communication theory to coordinate distributed heterogeneous agents, including human users, AI models, XR devices, and compute nodes. Additionally, we outline a distributed implementation model and discuss potential applications in immersive computing, scientific simulation, neurogaming, and cognitive digital twins. Finally, we define an experimental roadmap for validating cognitive-adaptive environments, establishing FCSTN as a foundational step toward next-generation intelligent virtual ecosystems.

Keywords: Fractal geometry, cognitive virtual environments, neurodigital interfaces, AI-generated worlds, coalition formation, space-time manifolds, extended reality, digital twins.

I. INTRODUCTION

The convergence of artificial intelligence, extended reality (XR) technologies, and neuroscience is fundamentally transforming the nature of human-computer interaction. Emerging neurodigital architectures aim to establish bidirectional communication between human cognition and artificial intelligence systems, moving beyond traditional input-output paradigms toward deeply integrated cognitive ecosystems [1]. Recent advances in brain-computer interface (BCI) technology have demonstrated that neural signals can be used to control virtual environments in real time, enabling new modalities of interaction that were previously confined to science fiction [2].

Simultaneously, theoretical physics provides a rich mathematical language for describing complex geometries and spatial structures. Solutions to Einstein's field equations produce non-Euclidean manifolds, exotic topologies, and dynamic curvature structures [3]. These mathematical models, when interpreted computationally, offer unprecedented possibilities for generating novel immersive environments whose spatial properties are derived from fundamental physics. The metric tensor formalism of general relativity, for instance, provides a natural framework for defining non-standard distance measures, curvature-based navigation, and topologically complex virtual spaces that differ fundamentally from conventional Euclidean worlds.

Previous work introduced the NeuroDigital Adaptive Network (NDAN), a decentralized infrastructure enabling interaction between human neural states and AI agents [4]. Such systems provide the foundation for immersive environments where cognition and computation become tightly integrated. The NDAN architecture includes cognitive synchronization protocols, AI-driven adaptive agents, and neural feedback mechanisms that collectively enable a closed-loop interaction paradigm between human users and virtual environments.

This paper proposes Fractal Cognitive Space-Time Networks (FCSTN), a unified framework that integrates relativistic geometry, fractal manifold generation, neurodigital interfaces, and coalition-based communication networks into a coherent architecture for AI-generated cognitive virtual environments. Within FCSTN, virtual universes are procedurally generated from mathematical structures such as Mandelbrot fractals and curved space-time metrics, while distributed devices and AI agents coordinate through coalition formation mechanisms derived from Device-to-Device (D2D) communication theory [5]. The resulting system forms a distributed cognitive universe where human neural signals, artificial intelligence, and computational infrastructure interact within dynamically evolving mathematical environments.

The main contributions of this paper are: (1) a comprehensive framework unifying fractal geometry, relativistic topology, neurodigital interfaces, and coalition-based networking; (2) a multi-layered architecture specification for FCSTN systems; (3) identification of key application domains including XR immersive worlds, neurogaming, cognitive digital twins, and scientific simulation; and (4) a detailed experimental roadmap outlining validation pathways for future implementation and evaluation.

II. THEORETICAL FOUNDATIONS

A. Relativistic Geometry as Computational Topology

General relativity describes space-time as a differentiable four-dimensional manifold endowed with a metric tensor that determines distances, angles, and curvature. The line element is expressed as:

$$ds^2 = g_{ij} dx^i dx^j$$

where g_{ij} is the metric tensor that encodes the local curvature of the manifold. In the context of virtual environment generation, this formalism provides a powerful tool: by defining different metric tensors, one can create virtual spaces with fundamentally different geometric properties. Flat Minkowski space yields conventional Euclidean navigation, while curved metrics produce environments where shortest paths between points follow geodesics rather than straight lines [3].

Exotic solutions to Einstein's field equations introduce topologically rich structures including curved geodesic trajectories, traversable wormhole geometries, and ring singularities. These features, when translated into virtual environment design, offer novel navigation mechanisms where users traverse curved paths through non-Euclidean spaces. Recent work in neural representations for computational general relativity has demonstrated that four-dimensional numerical solutions can be efficiently compressed and rendered using neural network architectures [6], suggesting that real-time visualization of curved space-time geometries is becoming computationally feasible.

For the FCSTN framework, we adopt a simplified computational interpretation: the metric tensor serves as a parameterizable function that defines spatial relationships in the virtual environment. AI agents can modulate metric parameters to reshape the environment dynamically, creating spaces that respond to cognitive input. This approach leverages the mathematical elegance of differential geometry while maintaining computational tractability for real-time applications.

B. Fractal Manifolds for Procedural Generation

Fractal dynamical systems provide a complementary mathematical structure for generating spatial complexity. The Mandelbrot iteration is defined as:

$$z_{n+1} = z_n^2 + c$$

where z and c are complex numbers. The resulting fractal set exhibits infinite structural detail and self-similarity across scales, making it an ideal candidate for procedural content generation in virtual environments. Recent applications in the gaming industry have demonstrated that fractal-based procedural generation can produce visually stunning and structurally complex worlds at runtime, with the Unreal Engine-based project "Chasing the Unseen" serving as a notable example of fractal mathematics powering explorable environments [7].

Within the FCSTN architecture, fractal fields define the spatial substrate of the virtual universe. Spatial properties are mapped from fractal parameters including iteration depth, escape velocity, fractal density gradients, and boundary

attractors. This mapping enables the generation of procedural universes with effectively infinite complexity, where each region of the environment exhibits unique geometric characteristics derived from the underlying fractal dynamics. The self-similar nature of fractals ensures visual coherence across scales, while the sensitivity to initial conditions provides environmental diversity.

C. Fractal-to-Virtual-World Generation Pipeline

The translation from mathematical fractal structures to immersive virtual environments follows a five-stage pipeline, as illustrated in Fig. 1. In Stage 1, fractal parameters (complex constants, iteration depth, escape radius) are selected and computed. Stage 2 generates a fractal field map that encodes spatial density, boundary attractors, and self-similarity layers across the virtual environment. Stage 3 maps the fractal field onto a differential geometric framework by computing metric tensor fields, curvature distributions, and geodesic paths. Stage 4 converts these mathematical structures into 3D mesh geometry with curved navigation surfaces and dynamic topology. Finally, Stage 5 renders the environment through real-time shaders with XR visualization capabilities and AI-adaptive visual parameters.

Fractal-to-Virtual-World Generation Pipeline

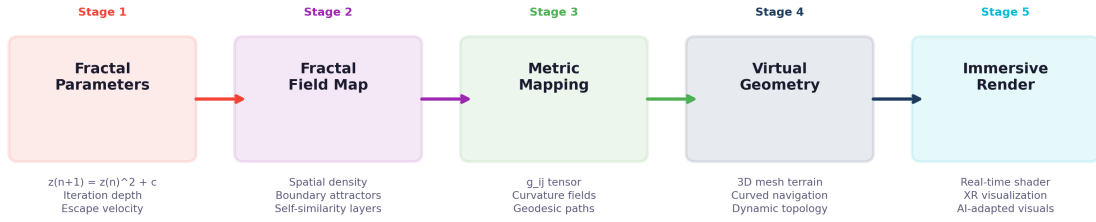


Fig. 1. Fractal-to-virtual-world generation pipeline showing the five stages from mathematical parameters to immersive rendering.

III. FCSTN FRAMEWORK ARCHITECTURE

A. System Overview

The FCSTN architecture is organized into four interconnected layers, each responsible for a distinct aspect of the cognitive virtual environment system. Fig. 2 presents the overall architecture, illustrating the flow of data and control between layers. Layer 1 (Cognitive Layer) encompasses human cognition, XR devices, and AI adaptive agents that serve as the primary sources of intent and interaction within the system. Layer 2 (Interface Layer) implements the NDAN protocol stack, including the Cognitive HoloChain synchronization protocol, BCI signal processing modules, AI adaptive agents, and a feedback controller. Layer 3 (Geometry Engine) contains the fractal space-time geometry engine with four core sub-components: fractal generation, metric computation, topology evolution, and procedural world-building. Layer 4 (Infrastructure Layer) provides the distributed computational substrate through D2D coalitions, edge computing nodes, cloud infrastructure, and sensor networks.

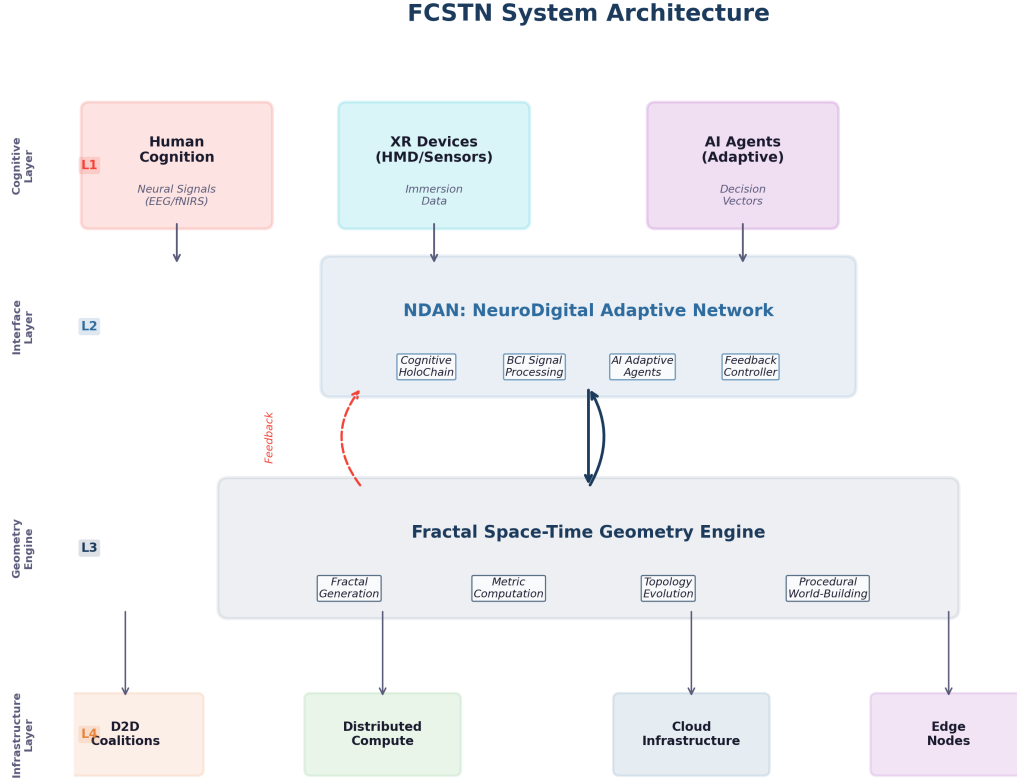


Fig. 2. FCSTN system architecture showing four interconnected layers: Cognitive Layer (L1), Interface Layer (L2), Geometry Engine (L3), and Infrastructure Layer (L4). Bidirectional feedback paths are shown between layers.

The key design principle of this architecture is bidirectional coupling: data flows both top-down (from cognitive intent to geometry generation) and bottom-up (from environmental state to sensory feedback). This creates a closed-loop system where the virtual environment continuously adapts to user cognition while simultaneously influencing user experience through generated spatial dynamics.

B. NeuroDigital Interface Layer (NDAN)

The NeuroDigital Adaptive Network (NDAN) serves as the critical interface between human cognition and the computational environment. Building on previous work [4], the NDAN layer within FCSTN is enhanced with several components specifically designed for cognitive virtual environment interaction. The Cognitive HoloChain Protocol provides decentralized synchronization of cognitive states across distributed users and AI agents, ensuring consistent environmental perception in multi-user scenarios. This protocol maintains a distributed ledger of cognitive interaction events, enabling persistent and reproducible virtual world states.

BCI Signal Processing modules translate raw neural signals (EEG, fNIRS) into environmental control parameters. Recent advances in neurodigital interfaces have demonstrated significant reductions in VR-induced motion sickness through vestibular integration approaches [8], suggesting that direct neural interfacing can improve both the fidelity and comfort of immersive experiences. The AI Adaptive Agents component employs machine learning models capable of interpreting neural signal patterns and translating them into environment modification commands. These agents learn individual user cognitive patterns over time, enabling personalized and increasingly responsive virtual environments.

C. AI-Driven Geometry Adaptation

A distinguishing feature of FCSTN is the role of artificial intelligence in dynamically modifying the geometric structure of the virtual environment. AI agents within the framework operate on multiple levels of the geometry generation pipeline. At the fractal level, AI models can adjust iteration parameters, color mappings, and boundary conditions to produce

environments that reflect cognitive states detected through the NDAN interface. At the metric level, neural networks can modulate curvature parameters, geodesic densities, and topological features to create spaces that feel intuitive yet mathematically rich.

This adaptive geometry approach creates environments that are fundamentally responsive to their inhabitants. A user exhibiting high cognitive engagement, as measured by EEG spectral features, might trigger the generation of more complex fractal regions with increased spatial detail. Conversely, signs of cognitive fatigue could prompt the AI to simplify the geometric complexity of the immediate environment, creating calmer navigational spaces. This bidirectional coupling between cognition and geometry represents a novel paradigm in human-computer interaction that goes beyond traditional adaptive interfaces.

D. Coalition-Based Distributed Infrastructure

Large-scale immersive environments require efficient and scalable communication infrastructure. FCSTN adopts a coalition formation approach derived from Device-to-Device (D2D) communication theory [5], [9]. In this model, heterogeneous devices including human users, AI agents, XR devices, and distributed compute nodes form dynamic coalitions to share computational resources and cognitive data streams. Let S denote a coalition of devices sharing communication resources. The coalition value is defined as:

$$v(S) = \text{Sum of } U_i \text{ for } i \text{ in } S$$

where U_i represents the communication utility or throughput of device i . The system objective is to maximize the total throughput across all coalitions subject to interference and power constraints. Within the FCSTN framework, this optimization extends beyond traditional communication metrics to include cognitive utility: coalitions are formed not only to optimize data transfer but also to maximize the quality of cognitive interaction within the virtual environment.

Fig. 3 illustrates the coalition formation model within FCSTN. Three types of coalitions emerge naturally: Cognitive Coalitions (connecting human users with AI agents and XR devices), Compute Coalitions (aggregating AI processing resources across edge and cloud nodes), and Infrastructure Coalitions (coordinating network resources for data-intensive fractal rendering tasks). The coalition formation algorithm follows a merge-and-split mechanism adapted from overlapping coalition game theory [10], enabling devices to dynamically join or leave coalitions based on changing computational demands and environmental conditions.

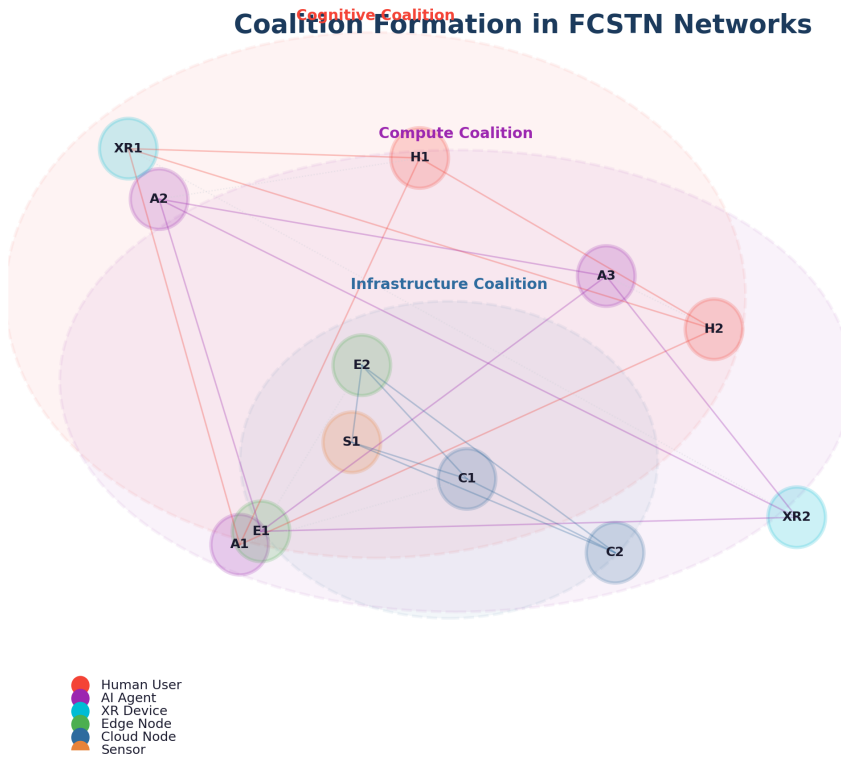


Fig. 3. Coalition formation model showing three coalition types: Cognitive, Compute, and Infrastructure. Nodes represent heterogeneous agents (H: human, A: AI, XR: device, E: edge, C: cloud, S: sensor).

IV. COGNITIVE FEEDBACK DYNAMICS

The core innovation of FCSTN lies in its bidirectional cognitive feedback architecture. As illustrated in Fig. 4, four principal components form a closed feedback loop: Human Cognition, AI Adaptive Agents, the Fractal World Geometry, and the NDAN Interface. Each component both influences and is influenced by the others, creating a continuously evolving system that adapts to its participants in real time.

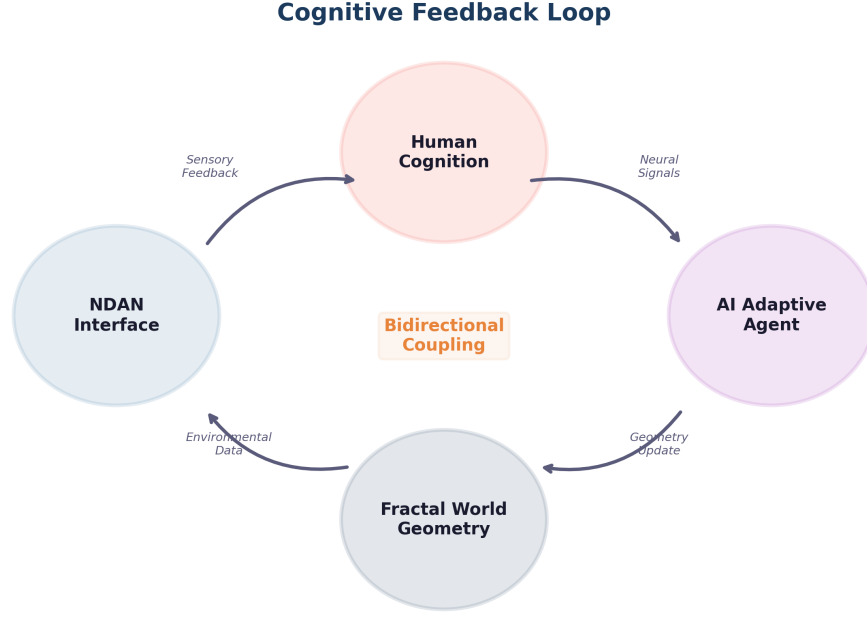


Fig. 4. Cognitive feedback loop showing bidirectional coupling between human cognition, AI adaptive agents, fractal world geometry, and the NDAN interface.

The feedback cycle operates as follows. Human cognition generates neural signals that are captured by BCI sensors and transmitted to the NDAN interface. The NDAN processes these signals and routes relevant cognitive data to AI adaptive agents. The AI agents interpret the cognitive state and compute geometry modification parameters that are applied to the fractal world engine. The modified virtual environment generates new sensory data (visual, auditory, haptic) that is rendered through XR devices and transmitted back to the user. This sensory feedback influences the user's subsequent cognitive state, completing the loop.

The temporal dynamics of this feedback loop are critical for user experience. Latency in any stage of the cycle can break the sense of immersion and cognitive coupling. Our architectural design targets sub-50ms round-trip latency for the complete feedback cycle, requiring optimized signal processing, efficient fractal computation, and low-latency communication infrastructure. The coalition-based networking approach supports this requirement by enabling local processing through edge nodes, reducing the need for cloud round-trips for time-critical operations.

V. APPLICATIONS AND USE CASES

A. Extended Reality Immersive Worlds

The FCSTN framework provides a natural platform for next-generation XR immersive worlds. Unlike conventional virtual environments that rely on pre-designed static geometry, FCSTN-generated worlds are procedurally created from mathematical structures and continuously adapted through cognitive feedback. This approach offers several advantages for immersive computing. First, the effectively infinite complexity of fractal generation ensures that users never exhaust the explorable content space, addressing a fundamental limitation of hand-crafted virtual worlds. Second, the integration of relativistic geometry enables novel navigation paradigms where users traverse curved geodesic paths, creating spatial experiences that are qualitatively different from conventional Euclidean VR environments.

Third, the cognitive feedback loop enables worlds that respond to collective human attention and emotional states. In a multi-user FCSTN environment, the aggregate cognitive state of all participants could influence the evolution of the shared virtual universe, creating emergent collective experiences that cannot be pre-programmed. This capability has significant implications for social VR platforms, collaborative workspaces, and educational environments where the spatial context

itself becomes a medium for communication and expression.

B. Neurogaming and Adaptive Entertainment

Neurogaming represents a natural application domain for FCSTN, as it combines neural interfacing, adaptive environments, and immersive technology into a cohesive entertainment experience. In an FCSTN-based neurogame, the player's cognitive and emotional states directly influence the game world: fractal environments morph in response to attention levels, stress indicators shape the difficulty and complexity of navigable spaces, and emotional arousal modulates visual aesthetics and environmental dynamics. This creates a deeply personalized gaming experience where the boundary between player and game world dissolves.

The procedural generation capabilities of FCSTN are particularly valuable for neurogaming, as they enable infinite content variety without manual asset creation. Each play session can generate a unique fractal world adapted to the player's current cognitive profile, ensuring that the experience remains fresh and engaging across repeated sessions. Furthermore, the AI-driven geometry adaptation can dynamically adjust the complexity and challenge level of the environment based on real-time cognitive metrics, creating a form of implicit difficulty scaling that responds to the player's mental state rather than explicit performance measures.

C. Cognitive Digital Twins

Digital twins have emerged as a powerful paradigm for creating virtual replicas of physical systems, with applications ranging from industrial manufacturing to urban planning [11]. The FCSTN framework extends the digital twin concept into the cognitive domain by enabling the creation of virtual environments that not only mirror physical systems but also replicate and respond to human cognitive states. A cognitive digital twin built on FCSTN would combine real-time sensor data from a physical system (through IoT integration), mathematical environment generation (through fractal geometry), and cognitive feedback (through the NDAN interface) to create a comprehensive virtual replica that operators can explore and interact with through XR devices.

Recent research on brain digital twins combining AI and extended reality [12] has demonstrated the potential of integrating neural data with immersive visualization for healthcare applications. The FCSTN framework provides the mathematical and architectural foundations for generalizing this concept to any domain where human cognitive interaction with complex systems is relevant, including healthcare, education, industrial training, and urban planning.

D. Scientific Simulation Environments

The mathematical rigor of the FCSTN framework makes it particularly suitable for scientific visualization and simulation. Relativistic geometry provides a natural framework for visualizing astrophysical phenomena such as black hole environments, gravitational lensing, and cosmological structure formation. Researchers could explore these environments interactively through XR devices, with the fractal generation engine providing the necessary visual complexity at multiple scales. Furthermore, the AI-driven adaptation could assist researchers by highlighting regions of interest, adjusting visualization parameters based on the scientific question being investigated, and facilitating collaborative exploration through shared cognitive virtual spaces.

The coalition-based infrastructure also supports distributed scientific collaboration, enabling researchers at different locations to simultaneously inhabit and interact with the same simulation environment. This capability transforms traditional visualization from a passive viewing experience into an active, embodied exploration of scientific data and mathematical structures.

Table 1. FCSTN application domains with key features, benefits, and validation metrics.

Application Domain	Key FCSTN Feature	Primary Benefit	Validation Metric
XR Immersive Worlds	Fractal generation + Cognitive feedback	Infinite content Personalized experience	User engagement Immersion score
Neurogaming	BCI integration + Adaptive geometry	Implicit difficulty Cognitive coupling	Session duration Cognitive load

Cognitive Digital Twins	NDAN + IoT + Fractal mapping	Cognitive-aware replication	Accuracy Response time
Scientific Simulation	Relativistic geometry + AI adaptation	Interactive exploration Multi-scale viz.	Visualization fidelity Collaboration index

VI. EXPERIMENTAL ROADMAP

To validate the FCSTN framework and establish its practical viability, we outline a four-phase experimental roadmap spanning approximately two years, as illustrated in Fig. 5. This roadmap is designed to progressively increase system complexity while establishing quantitative benchmarks at each stage. We emphasize that this roadmap outlines prototype architecture and future experimental validation pathways, providing a structured approach to transitioning the FCSTN framework from theoretical concept to implemented system.

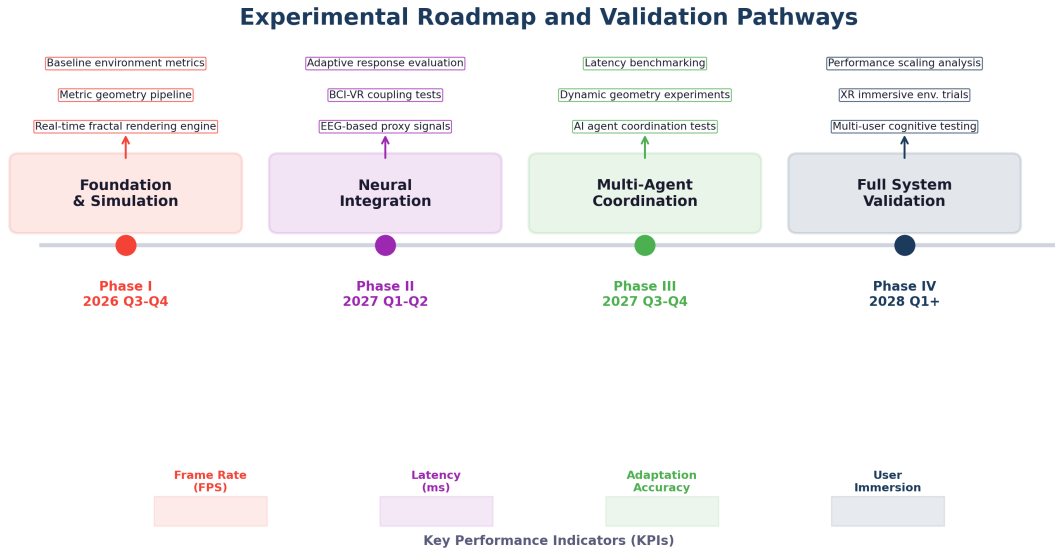


Fig. 5. Four-phase experimental roadmap for FCSTN validation, from foundation simulation (Phase I) through full system validation (Phase IV), with key performance indicators.

A. Phase I: Foundation and Simulation (2026 Q3-Q4)

The first phase focuses on establishing the core computational infrastructure for FCSTN. The primary deliverable is a real-time fractal rendering engine capable of generating and displaying complex fractal-based virtual environments at interactive frame rates (target: 60 FPS minimum). This engine will be built using commercially available game engines (Unreal Engine 5 or Unity) with custom compute shaders for fractal computation. The metric geometry pipeline will be implemented as a separate module that maps fractal parameters onto differential geometric structures, with baseline measurements of computation time, memory usage, and rendering quality established during this phase.

B. Phase II: Neural Integration (2027 Q1-Q2)

The second phase introduces neural signal integration into the system. EEG-based proxy signals will serve as the primary input modality, with open-source EEG datasets used for initial testing before progressing to live BCI hardware. Key experiments include: (1) BCI-VR coupling tests measuring the latency and accuracy of neural-to-environment signal translation, (2) adaptive response evaluation comparing environment adaptation quality with and without neural input, and (3) user comfort studies assessing the impact of neural feedback on immersion and motion sickness. Performance targets include sub-100ms neural signal processing latency and environment adaptation accuracy exceeding 70% correlation with input cognitive features.

C. Phase III: Multi-Agent Coordination (2027 Q3-Q4)

The third phase scales the system to support multiple AI agents operating simultaneously within the same virtual environment. Experiments will evaluate: (1) AI agent coordination in dynamic fractal geometries, measuring convergence time and resource utilization efficiency; (2) dynamic geometry experiments testing the system's ability to maintain environmental coherence under multiple simultaneous modification requests; and (3) latency benchmarking for the complete feedback cycle under varying network conditions and coalition configurations. The coalition formation algorithm will be implemented and evaluated using both simulation and emulation on a small-scale testbed.

D. Phase IV: Full System Validation (2028 Q1+)

The final phase integrates all components into a complete FCSTN system and conducts comprehensive validation experiments. Multi-user cognitive testing will evaluate the system's performance with multiple simultaneous human participants, measuring environmental coherence, cognitive feedback fidelity, and collective experience quality. XR immersive environment trials will assess the system's performance across different XR hardware platforms, including standalone VR headsets and room-scale AR systems. Performance scaling analysis will characterize the system's behavior as the number of users, AI agents, and environmental complexity increases, establishing the practical scalability limits of the FCSTN architecture.

Table 2. Key performance indicators with phase-specific targets and measurement methods.

Key Performance Indicator	Target (Phase II)	Target (Phase IV)	Measurement Method
Render Frame Rate	60 FPS	90 FPS	Built-in profiler
Neural Processing Latency	< 100 ms	< 50 ms	Signal round-trip timing
Environment Adaptation Accuracy	> 70%	> 85%	Correlation analysis
Coalition Formation Convergence	< 5 s	< 2 s	Algorithm iteration count
Multi-User Coherence	N/A	> 95%	State synchronization tests
User Immersion Score	> 3.5/5	> 4.2/5	Standardized UX questionnaire

VII. DISCUSSION AND LIMITATIONS

The FCSTN framework represents an ambitious attempt to unify concepts from theoretical physics, fractal mathematics, artificial intelligence, and neurodigital interfacing into a coherent architecture for cognitive virtual environments. While the framework offers significant conceptual novelty, several limitations and challenges must be acknowledged.

First, the computational cost of real-time fractal generation and metric tensor computation remains a significant challenge. Although modern GPU architectures and compute shaders have dramatically improved the feasibility of real-time fractal rendering, the additional overhead of metric computation, AI-driven adaptation, and neural signal processing creates a demanding computational pipeline. The coalition-based distributed architecture partially addresses this by distributing computation across edge and cloud nodes, but network latency and bandwidth constraints may limit the achievable performance in geographically distributed deployments.

Second, the current framework relies on simplified mathematical models that may not capture the full richness of the physical theories they draw upon. The computational interpretation of relativistic geometry adopted in this paper abstracts away many of the complexities of general relativity, including the coupling between geometry and matter-energy distributions. Future work should explore more sophisticated implementations that incorporate additional physical phenomena, such as time dilation effects and gravitational wave dynamics, into the virtual environment generation pipeline.

Third, the neurodigital interface component faces practical challenges related to signal quality, individual variability in neural responses, and the current limitations of consumer-grade BCI hardware. While research-grade EEG systems can

provide sufficient signal quality for cognitive state estimation, the transition to consumer-grade devices introduces noise and variability that may degrade the quality of the cognitive feedback loop. The framework's AI adaptive agents are designed to mitigate this through personalized learning and noise-robust signal processing, but the extent to which these approaches can compensate for hardware limitations remains an open question.

Finally, ethical considerations surrounding neurodigital interfaces must be carefully addressed. Systems that directly measure and respond to neural activity raise important questions about cognitive privacy, informed consent, and the potential for manipulation of user experience. The FCSTN framework incorporates these concerns into its design through the NDAN's decentralized architecture, which provides users with control over their cognitive data, and the coalition formation model, which ensures that no single entity has unilateral control over the virtual environment.

VIII. CONCLUSION

This paper has presented Fractal Cognitive Space-Time Networks (FCSTN), a comprehensive framework for AI-generated cognitive virtual environments that unifies concepts from fractal mathematics, relativistic geometry, neurodigital interfacing, and coalition-based distributed networking. The four-layer architecture provides a structured approach to building virtual universes that are procedurally generated from mathematical structures, continuously adapted through AI-driven feedback, and grounded in human cognitive states through neurodigital interfaces.

The framework identifies four key application domains, XR immersive worlds, neurogaming, cognitive digital twins, and scientific simulation, each leveraging different aspects of the FCSTN architecture to address specific use cases. The proposed experimental roadmap provides a structured pathway for validating the framework through progressive system integration, with clearly defined performance targets and measurement methodologies across four phases.

FCSTN represents a foundational step toward next-generation intelligent virtual ecosystems where the boundary between human cognition and computational environments becomes increasingly fluid. As AI, XR, and neurodigital technologies continue to mature, frameworks like FCSTN will become essential for building the cognitive virtual environments that define the future of human-computer interaction and human-AI symbiosis. Future work will focus on implementing the prototype system described in the experimental roadmap and conducting empirical validation of the cognitive feedback mechanisms that constitute the core innovation of this framework.

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